

ÖMER SAFA ÇAVUŞ

MANUFACTURING ENGINEER

Istanbul, Turkey
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EDUCATION

- Sabanci University, Ph.D. in Manufacturing Engineering** 2020 – present, Istanbul
PhD Thesis: Planning and analysis of laser beam powder bed fusion processes supervised by Prof. Dr. Bahattin Koç
- Istanbul Technical University, B.Sc. in Civil Engineering** 2014 – 2020, Istanbul
Graduation Project: Repairing, protecting and strengthening ageing, damaged or overloaded concrete, and masonry structures with FRCM supervised by Assoc. Prof. Dr. Bekir Yılmaz Pekmezci
- Hamburg University of Technology, B.Sc. in Civil Engineering** 2018, Hamburg
Erasmus+ Exchange Program

PROFESSIONAL EXPERIENCE

- Machina3D , R&D Consultant** 2024 – 2026
Additive Manufacturing: Provided technical consulting on end-to-end process design, including part-scale thermal analysis, support structure engineering, and geometry optimization based on Design for Additive Manufacturing (DfAM) principles.
Casting: Guided the development of in-house software for automated mold generation and advised on heat treatment recipes and advanced material characterization methods, with a primary focus on aluminum alloys.
- Sabanci University Integrated Manufacturing Technologies Research and Application Center (SU-IMC) , Researcher** 2020 – present
- Project 1: New Generation 3D Printing Manufacturing Technologies Platform. Conducting research to develop next-generation additive manufacturing technologies and produce high value-added products for the aerospace and defense industries. (Partners: TUBITAK, Sabanci University, ROKETSAN)* 2023 – present
- Project 2: Development and Experimental Validation of Innovative Topology Optimization Algorithms Supported by Numerical Models. Developed in-house scanning strategy algorithms and computational environments utilizing **computer graphics** and **parametric modeling** techniques. Developed in-house **multi-physics** solvers and implemented process simulations across different scales. Conducted comprehensive material **characterization** and **mechanical testing** to validate numerical models. (Partners: TUBITAK, Sabanci University, TEI)* 2020 – 2023
- Project 3: Wear and Corrosion Protection Coatings with Thermal Management Using Metal Matrix Composites (Partners: TUBITAK, Sabanci University)* 2025 – 2027
- Sabanci University, Teaching Assistant** 2020 – present
- Course 1: Introduction to Computer-Aided Drafting and Solid Modeling with SolidWorks * 2020 – 2022
- Course 2: Manufacturing Processes I * 2022 – 2023
- Course 3: Additive Manufacturing * 2024 – present

JOURNAL PAPERS

- Khalilvandi Behrouzfar, S., **Cavus, O. S.**, & Koc, B. (2026). A parametric heat source formulation for arbitrary laser beam shaping: application to core-ring profiles and their transient thermal history in PBF-LB. *Submitted to International Journal of Extreme Manufacturing*.
- Cavus, O. S.**, Khalilvandi Behrouzfar, S., & Koc, B. (2026). Prediction of cooling rates based on integrated scanning strategies with sophisticated parametric modeling-assisted simulation for additive manufacturing. *Progress in Additive Manufacturing*. 
- Cavus, O. S.**, Khalilvandi Behrouzfar, S., Javidrad, H. R., & Koc, B. (2026). Continuous parametric scanning with thermal modeling to minimize stitch zone defects in laser beam powder bed fusion validated by multi-scale X-ray CT *Additive Manufacturing*. 
- Kendibilir, A., Khalilvandi Behrouzfar, S., **Cavus, O. S.**, Lahe Motlagh, P., Koc, B., & Kefal, A. (2026). Enhancing mechanical performance of metal additively manufactured structures via peridynamic topology optimization and thermo-mechanical process simulation with experimental validation. *Computers & Structures*. 
- Khalilvandi Behrouzfar, S., Kendibilir, A., **Cavus, O. S.**, Lahe Motlagh, P., Koc, B., & Kefal, A. (2025). Minimizing thermally induced residual stresses in metal additive manufacturing through peridynamics topology optimization. *Engineering with Computers*. 
- Cavus, O. S.**, Khalilvandi Behrouzfar, S., & Koc, B. (2024). Functional island scanning strategy for powder bed fusion. *Journal of Additive Manufacturing Technologies*. 
- Kendibilir, A., Polat, M. F., **Cavus, O. S.**, Behrouzfar, S. K., Motlagh, P. L., Koc, B., & Kefal, A. (2022). Combination of peri-

dynamics and genetic algorithm based topology optimization methods for AM-friendly designs. *Journal of Additive Manufacturing Technologies*, 2(1), 704-704. 

CONFERENCE PAPERS

1. **Cavus, O. S.**, Khalilvandi Behrouzyar, S., Javidrad, H. R., & Koc, B. (2026). Investigating the role of scanning strategies on porosity formation and mitigation in stitch regions via thermal modeling and X-Ray CT. *Paper presented at the 6th AMCTURKEY Additive Manufacturing Conference, Antalya, Turkey.*
2. Ozyilmaz, P., **Cavus, O. S.**, Khalilvandi Behrouzyar, S., Erkul, I., & Koc, B. (2026). Assessing the limits of in-situ back-scatter electron imaging in narrow electron beam powder bed fusion process windows using X-Ray CT. *Paper presented at the 6th AMCTURKEY Additive Manufacturing Conference, Antalya, Turkey.*
3. **Cavus, O. S.**, Khalilvandi Behrouzyar, S., & Koc, B. (2025). Transient and spatial evolution of temperature gradients and solidification rates in laser beam powder bed fusion. *Paper presented at the 5th AMCTURKEY Additive Manufacturing Conference, Antalya, Turkey.* 
4. Khalilvandi Behrouzyar, S., **Cavus, O. S.**, & Koc, B. (2025). A comparative study of single and multi-laser beam effects on thermal gradient and solidification rate in LB-PBF. *Paper presented at the 5th AMCTURKEY Additive Manufacturing Conference, Antalya, Turkey.* 
5. **Cavus, O. S.**, Khalilvandi Behrouzyar, S., & Koc, B. (2024). Design thermal analysis and optimization of the novel geometry based scan strategies. *Paper presented at the 4th AMCTURKEY Additive Manufacturing Conference, Antalya, Turkey.* 

PROJECTS UNDER REVIEW

1. **Horizon Europe Framework Programme, Researcher, Principal Investigator: Prof. Dr. Bahattin Koç**
Project: Next-generation Engineering for Optimized Lifecycle of Critical Materials Supported by AI and Additive Manufacturing to Promote Re-standardization and Circularity
(This proposal will be resubmitted in the upcoming call following the initial review.)
2. **Marie Skłodowska-Curie Actions, Researcher, Principal Investigator: Prof. Dr. Bahattin Koç**
Project: Laser-Based Digital Manufacturing for Energy Efficient Processes and Products
3. **TÜBİTAK-GSRI, Researcher, Principal Investigator: Prof. Dr. Bahattin Koç**
Project: Integrated Additive Manufactured Sensing System for Marine Applications

ENGINEERING SKILLS

- **Additive Manufacturing:** Materialise Magics, Simufact Additive, Netfabb, PBF-LB: EOS M 290, PBF-EB: Freemelt ONE, Cold Spray: Dycomet Titomic PCS-100
- **Numerical Simulation:** MSC Marc, ANSYS Workbench, User Defined Subroutines, Flow-3D AM
- **Material Characterization and Microstructure:** X-Ray Diffraction Analysis (XRD), X-Ray Computed Tomography (X-Ray CT), Scanning Electron Microscope (SEM), Optical Microscope, SPPARKS Kinetic Monte Carlo Simulator
- **Programming:** MATLAB, Python, C#, Fortran, L^AT_EX
- **Design:** Parametric Modelling, Rhinoceros/Grasshopper, SolidWorks
- **Topology Optimized Based Design:** MSC Apex, Fusion 360, ANSYS Workbench, nTop
- **Other:** AI, Machine Learning & Statistics, Mechanical Behaviour of Materials, Manufacturing Fundamentals, Engineering Optimization

CERTIFICATES & TRAININGS

DIMAP Project Trainings

DIMAP  (Direct Digital Manufacturing Platform), co-financed by the European Union and the Republic of Turkey. The project aims to facilitate the implementation of digital production and additive manufacturing methods, particularly in the aviation and automotive sectors.

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| • Directed Energy Deposition & Post Processes , Politecnico di Milano, by Prof. Ali Gökhan Demir | 2025 |
| • Practical Training on Cold Spray , Titomic, by Ferdy Touwen | 2025 |
| • Theory of Cold Spray , Politecnico di Milano, by Prof. Sara Bagherifard | 2025 |
| • Scanning Electron Microscopy , Zeiss, by Volkan Erkut | 2025 |
| • Digital Simulation , Politecnico di Milano, by Prof. Marco Grasso | 2025 |
| • Electron Beam Melting , Freemelt, by Camilo Medina Viramontes | 2025 |
| • X-ray Diffraction (XRD) , Anka Analitik, by Feda Rahal | 2025 |

AWARDS

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| • Performance Scholarships , TUBITAK | 2024 |
| • Project Scholarships , TUBITAK | 2020 |
| • High Honour List , Istanbul Technical University | 2018 |
| • Grant for Erasmus+ Exchange Students , European Union | 2018 |
| • A2 German Language Certificate , Goethe Institut | 2017 |
| • Certificate of Appreciation , UNICEF | 2017 |

LANGUAGES

Turkish (Native), English (Fluent), German (Elementary)

INTERESTS

Skiing, Scuba Diving, Travelling

REFERENCES

Prof. Dr. B. Koç
Sabanci University

Assoc. Prof. A. Kefal
Sabanci University

Selim Dinçer
Machina3D

Dr. Utku Seven
Sabanci University